

LE CHUNG HIEU

DIGITAL IC DESIGN INTERN

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PERSONAL STATEMENT

Final-year Electronics and Telecommunications Engineering student at Ho Chi Minh City University of Technology (HCMUT), pursuing a career in Digital IC Design with strong interests in Design for Testability (DFT), Static Timing Analysis (STA), and ASIC implementation flows. Experienced in RTL design, FPGA prototyping, and functional verification using Verilog/SystemVerilog. Hands-on experience with pipelined processor design, cryptographic accelerators, DSP architectures, and FPGA deployment. Familiar with timing concepts including setup/hold analysis, clock domain crossing, timing constraints, and pipeline optimization. Seeking an internship position in DFT/STA where I can contribute my digital design background while expanding my expertise in timing closure, testability, ATPG, scan architecture, and advanced VLSI implementation methodologies.

EDUCATION

Ho Chi Minh City University of Technology (VNU-HCMUT)	2022 – Current
Final-year student	GPA: 7.1/10
Faculty: Electrical and Electronics Engineering	Major: Electronics and Telecommunications Engineering

LANGUAGES

TOEIC (LISTENING/READING): 790/990

PROJECTS

COMPUTER ARCHITECTURE (INSTRUCTOR: ASSC. PROF. DR. TRAN HOANG LINH)

Ho Chi Minh City University of Technology	Oct 2025 – Dec 2025
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Technologies: SystemVerilog, Cadence Xcelium, Quartus, DE2-115 Altera FPGA Kit.

Description: Designed and implemented a 32-bit RV32I RISC-V processor in SystemVerilog, including Program Counter (PC), Register File, ALU, Immediate Generator, Control Unit, and memory subsystems. Extended the design from a single-cycle architecture to a 5-stage pipelined architecture (IF, ID, EX, MEM, WB) with hazard detection and forwarding mechanisms. Verified functionality using Cadence Xcelium through RTL simulation and waveform debugging. Synthesized and deployed the processor on a DE2-115 FPGA platform, successfully executing application programs such as a stopwatch. Analyzed critical datapaths, pipeline latency, forwarding paths, and timing behavior under different clock constraints, gaining practical understanding of setup/hold timing considerations and timing closure challenges in pipelined digital systems.

CRYSTALS-DILITHIUM PQC SIGNATURE - FPGA IMPLEMENTATION (INSTRUCTOR: ASSC. PROF. DR. TRAN HOANG LINH)

Ho Chi Minh City University of Technology	Oct 2025 – Dec 2025
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Technologies: Verilog, VHDL, Vivado, Quartus, ModelSim.

Description: Implemented major hardware blocks of the CRYSTALS-Dilithium post-quantum digital signature algorithm targeting FPGA platforms. Designed Number Theoretic Transform (NTT) and Inverse NTT engines, butterfly processing units, modular reduction circuits, twiddle-factor memory architecture, and address generation logic. Optimized datapath architecture to improve computational efficiency while reducing hardware resource utilization. Verified functionality through RTL simulation, waveform debugging, and comparison against software-generated reference vectors. Performed FPGA synthesis and timing evaluation using Vivado and Quartus, analyzing critical arithmetic paths and memory-access latency.

FPGA-BASED AES-CTR ENCRYPTION CORE (INSTRUCTOR: M.SC. NGUYEN TUAN HUNG)

Ho Chi Minh City University of Technology

Jun 2025 – Aug 2025

Technologies: SystemVerilog, Vivado, Quartus.

Description: Designed and implemented a hardware encryption accelerator based on AES operating in Counter (CTR) mode, supporting AES-128, AES-192, and AES-256 key configurations. Developed a modular RTL architecture consisting of Key Expansion, Cipher Engine, CTR Controller, Counter Generator, and encryption datapath modules. Optimized control and datapath organization to improve design scalability and throughput efficiency. Performed functional verification using SystemVerilog testbenches, waveform analysis, and AES standard test vectors. Successfully synthesized and deployed the design on a DE1-SoC FPGA platform while evaluating datapath timing behavior, resource utilization, and hardware performance.

DIGITAL SIGNAL PROCESSING ON FPGA (INSTRUCTOR: M.SC. NGUYEN TUAN HUNG)

Ho Chi Minh City University of Technology

Jan 2025 – May 2025

Technologies: SystemVerilog, Quartus, ModelSim.

Description: Designed and implemented real-time digital signal processing modules on FPGA targeting audio applications. Developed FIR and IIR filter architectures including coefficient storage, multiply-accumulate datapaths, and control logic. Integrated an I2C-based audio codec interface for real-time audio acquisition and playback. Performed RTL simulation, functional verification, and FPGA implementation using Quartus and ModelSim. Analyzed timing reports, datapath delays, and clock-domain interactions to ensure timing correctness under real-time processing requirements while strengthening understanding of setup/hold constraints and synchronous digital design methodologies.

SKILLS

- **HDL Languages:** SystemVerilog, Verilog, VHDL.
- **Digital Design:** RTL Design, FSM Design, Pipeline Architecture, Datapath Design, RISC-V Processor Design, DSP Hardware Design, Cryptographic Accelerator Design.
- **Verification:** Testbench Development, Self-checking Verification, Functional Simulation, Waveform Debugging, Coverage-oriented Verification.
- **Timing Analysis:** Setup/Hold Analysis, Critical Path Identification, Timing Constraints, Clocking Concepts, Basic Static Timing Analysis (STA).
- **DFT Knowledge:** Scan Chain Fundamentals, Controllability/Observability Concepts, Fault Models (Stuck-at Fault), ATPG Fundamentals.
- **EDA Tools:** Cadence Xcelium, ModelSim, Vivado, Quartus Prime, Visual Studio Code.
- **FPGA Platforms:** DE1-SoC, DE2-115, DE10.
- **Programming:** Python, MATLAB, C/C++.
- **Soft Skills:** Teamwork, Technical Presentation, Problem Solving, Documentation.